

2.0 EXTERNAL CONNECTIONS

2.1 Oscillator

The ENC28J60 is designed to operate at 25 MHz with a crystal connected to the OSC1 and OSC2 pins. The ENC28J60 design requires the use of a parallel cut crystal. Use of a series cut crystal may give a frequency out of the crystal manufacturer specifications. A typical oscillator circuit is shown in Figure 2-1.

The ENC28J60 may also be driven by an external clock source connected to the OSC1 pin as shown in Figure 2-2.

FIGURE 2-1: CRYSTAL OSCILLATOR OPERATION

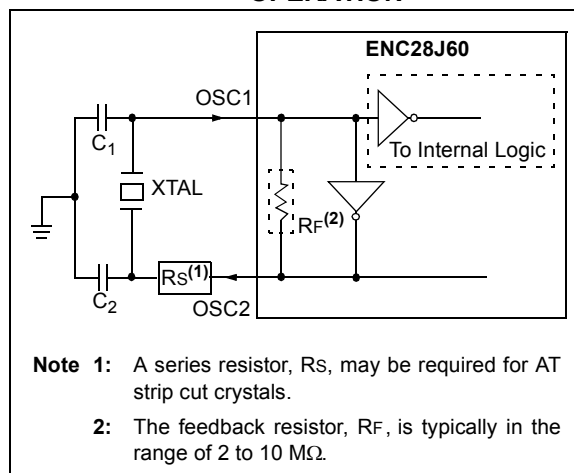
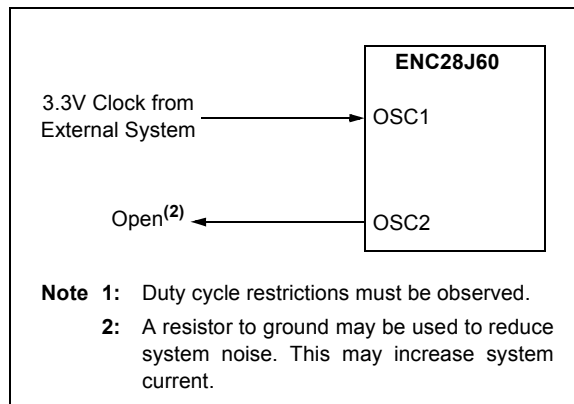


FIGURE 2-2: EXTERNAL CLOCK SOURCE⁽¹⁾



2.2 Oscillator Start-up Timer

The ENC28J60 contains an Oscillator Start-up Timer (OST) to ensure that the oscillator and integrated PHY have stabilized before use. The OST does not expire until 7500 OSC1 clock cycles (300 μ s) pass after Power-on Reset or wake-up from Power-Down mode occurs. During the delay, all Ethernet registers and buffer memory may still be read and written to through the SPI bus. However, software should not attempt to transmit any packets (set ECON1.TXRTS), enable reception of packets (set ECON1.RXEN) or access any MAC, MII or PHY registers during this period.

When the OST expires, the CLKRDY bit in the ESTAT register will be set. The application software should poll this bit as necessary to determine when normal device operation can begin.

Note: After a Power-on Reset, or the ENC28J60 is removed from Power-Down mode, the CLKRDY bit must be polled before transmitting packets, enabling packet reception or accessing any MAC, MII or PHY registers.

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2.3 CLKOUT Pin

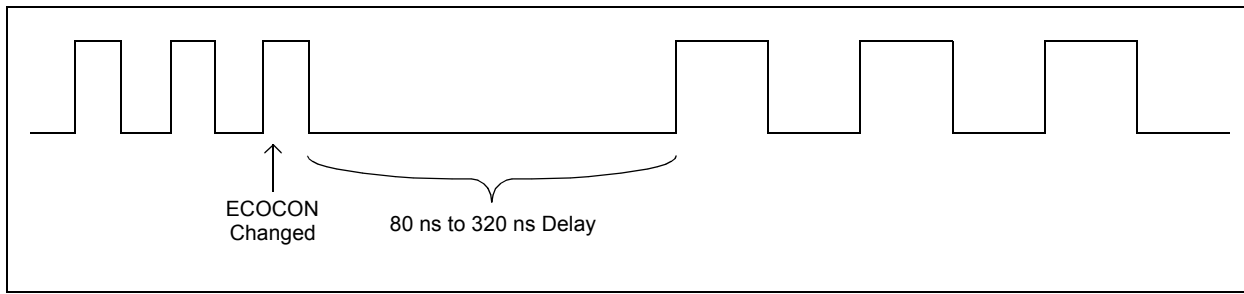
The clock out pin is provided to the system designer for use as the host controller clock or as a clock source for other devices in the system. The CLKOUT has an internal prescaler which can divide the output by 1, 2, 3, 4 or 8. The CLKOUT function is enabled and the prescaler is selected via the ECOCON register (Register 2-1).

To create a clean clock signal, the CLKOUT pin is held low for a period when power is first applied. After the Power-on Reset ends, the OST will begin counting. When the OST expires, the CLKOUT pin will begin outputting its default frequency of 6.25 MHz (main clock divided by 4). At any future time that the ENC28J60 is reset by software or the $\overline{\text{RESET}}$ pin, the CLKOUT function will not be altered (ECOCON will not change

value). Additionally, Power-Down mode may be entered and the CLKOUT function will continue to operate. When Power-Down mode is cancelled, the OST will be reset but the CLKOUT function will continue. When the CLKOUT function is disabled (ECOCON = 0), the CLKOUT pin is driven low.

The CLKOUT function is designed to ensure that minimum timings are preserved when the CLKOUT pin function is enabled, disabled or the prescaler value is changed. No high or low pulses will be outputted which exceed the frequency specified by the ECOCON configuration. However, when switching frequencies, a delay between two and eight OSC1 clock periods will occur where no clock pulses will be produced (see Figure 2-3). During this period, CLKOUT will be held low.

FIGURE 2-3: CLKOUT TRANSITION



REGISTER 2-1: ECOCON: CLOCK OUTPUT CONTROL REGISTER

U-0	U-0	U-0	U-0	U-0	R/W-1	R/W-0	R/W-0
—	—	—	—	—	COCON2	COCON1	COCON0
bit 7					bit 0		

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-3 **Unimplemented:** Read as '0'

bit 2-0 **COCON2:COCON0:** Clock Output Configuration bits

11x = Reserved for factory test. Do not use. Glitch prevention not assured.

101 = CLKOUT outputs main clock divided by 8 (3.125 MHz)

100 = CLKOUT outputs main clock divided by 4 (6.25 MHz)

011 = CLKOUT outputs main clock divided by 3 (8.333333 MHz)

010 = CLKOUT outputs main clock divided by 2 (12.5 MHz)

001 = CLKOUT outputs main clock divided by 1 (25 MHz)

000 = CLKOUT is disabled. The pin is driven low.

2.4 Magnetics, Termination and Other External Components

To complete the Ethernet interface, the ENC28J60 requires several standard components to be installed externally. These components should be connected as shown in Figure 2-4.

The internal analog circuitry in the PHY module requires that an external 2.32 k Ω , 1% resistor be attached from RBIAS to ground. The resistor influences the TPOUT+/- signal amplitude. The resistor should be placed as close as possible to the chip with no immediately adjacent signal traces to prevent noise capacitively coupling into the pin and affecting the transmit behavior. It is recommended that the resistor be a surface mount type.

Some of the device's digital logic operates at a nominal 2.5V. An on-chip voltage regulator is incorporated to generate this voltage. The only external component required is an external filter capacitor, connected from VCAP to ground. The capacitor must have low equivalent series resistance (ESR), with a typical value of 10 μ F, and a minimum value of 1 μ F. The internal regulator is not designed to drive external loads.

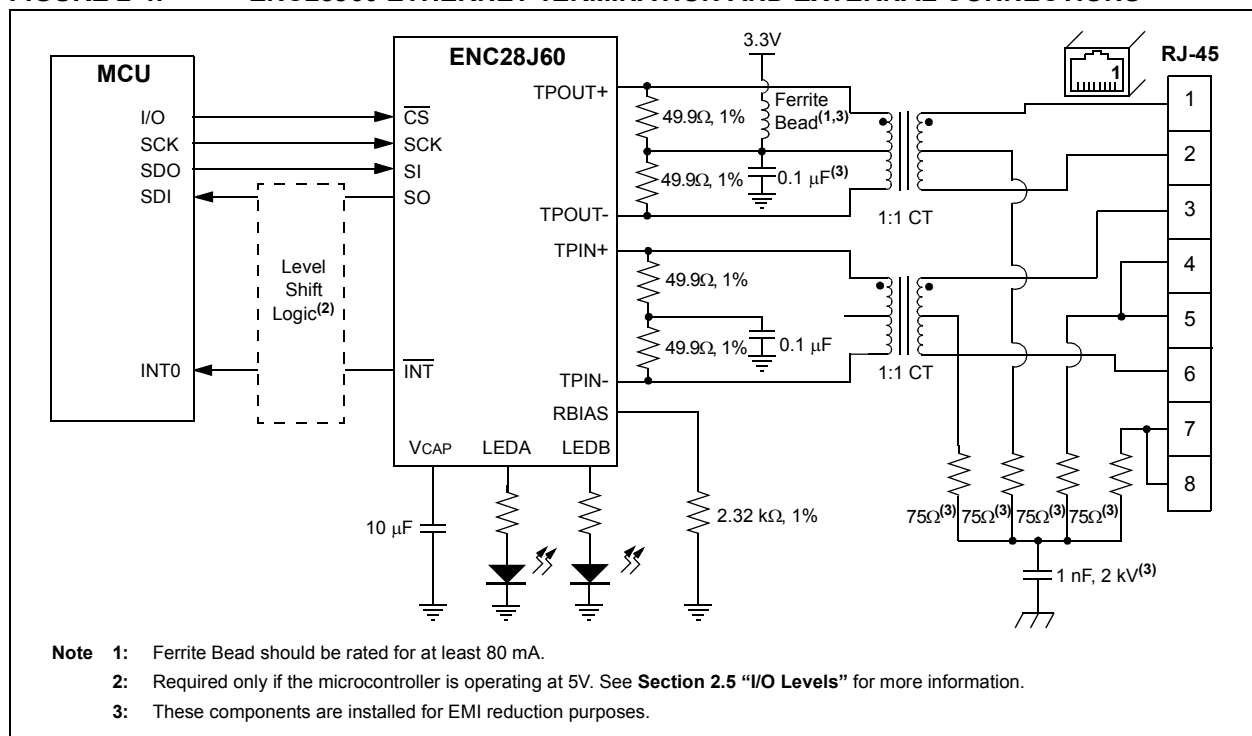
On the TPIN+/TPIN- and TPOUT+/TPOUT- pins, 1:1 center tapped pulse transformers, rated for Ethernet operations, are required. When the Ethernet module is enabled, current is continually sunk through both TPOUT pins. When the PHY is actively transmitting, a differential voltage is created on the Ethernet cable by varying the relative current sunk by TPOUT+ compared to TPOUT-.

A common-mode choke on the TPOUT interface, placed between the TPOUT pins and the Ethernet transformer (not shown), is not recommended. If a common-mode choke is used to reduce EMI emissions, it should be placed between the Ethernet transformer and pins 1 and 2 of the RJ-45 connector. Many Ethernet transformer modules include common-mode chokes inside the same device package. The transformers should have at least the isolation rating specified in Table 16-5 to protect against static voltages and meet IEEE 802.3 isolation requirements (see **Section 16.0 "Electrical Characteristics"** for specific transformer requirements). Both transmit and receive interfaces additionally require two resistors and a capacitor to properly terminate the transmission line, minimizing signal reflections.

All power supply pins must be externally connected to the same power source. Similarly, all ground references must be externally connected to the same ground node. Each VDD and VSS pin pair should have a 0.1 μ F ceramic bypass capacitor (not shown in the schematic) placed as close to the pins as possible.

Since relatively high currents are necessary to operate the twisted-pair interface, all wires should be kept as short as possible. Reasonable wire widths should be used on power wires to reduce resistive loss. If the differential data lines cannot be kept short, they should be routed in such a way as to have a 100 Ω characteristic impedance.

FIGURE 2-4: ENC28J60 ETHERNET TERMINATION AND EXTERNAL CONNECTIONS



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2.5 I/O Levels

The ENC28J60 is a 3.3V part; however, it was designed to be easily integrated into 5V systems. The SPI CS, SCK and SI inputs, as well as the RESET pin, are all 5V tolerant. On the other hand, if the host controller is operated at 5V, it quite likely will not be within specifications when its SPI and interrupt inputs are driven by the 3.3V CMOS outputs on the ENC28J60. A unidirectional level translator would be necessary.

An economical 74HCT08 (quad AND gate), 74ACT125 (quad 3-state buffer) or many other 5V CMOS chips with TTL level input buffers may be used to provide the necessary level shifting. The use of 3-state buffers permits easy integration into systems which share the SPI bus with other devices. Figure 2-5 and Figure 2-6 show example translation schemes.

FIGURE 2-5: LEVEL SHIFTING USING AND GATES

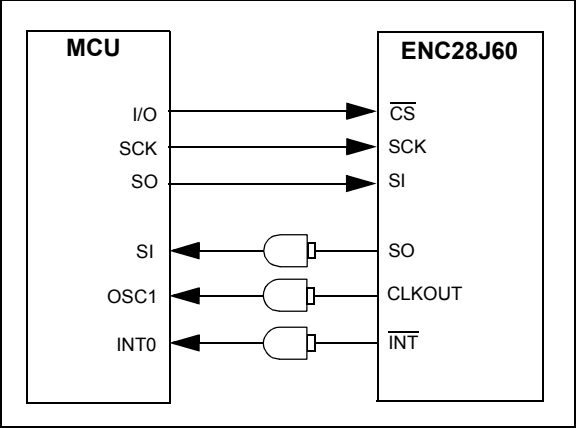
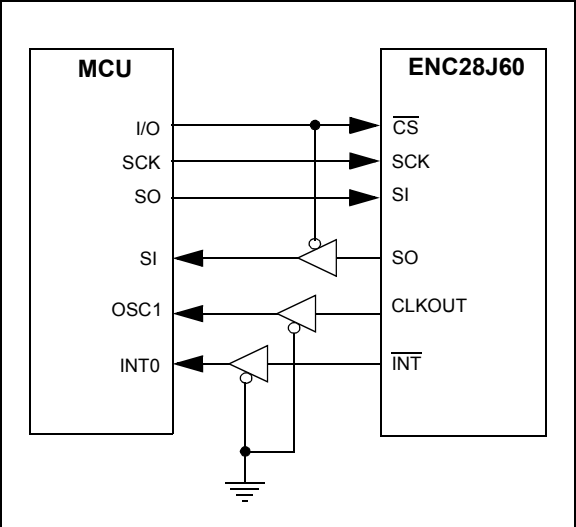


FIGURE 2-6: LEVEL SHIFTING USING 3-STATE BUFFERS

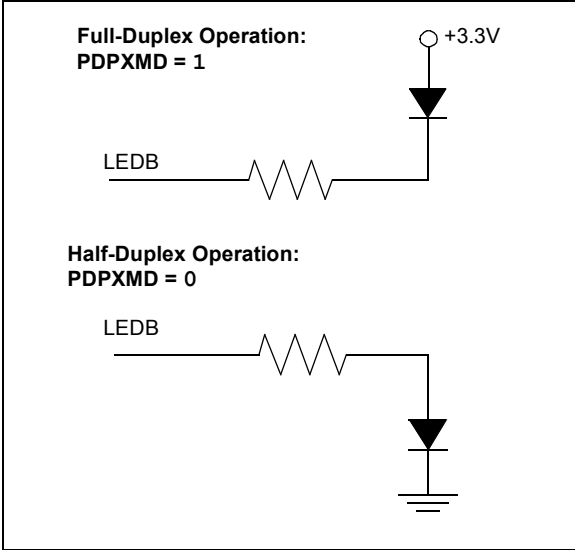


2.6 LED Configuration

The LEDA and LEDB pins support automatic polarity detection on Reset. The LEDs can be connected such that the pin must source current to turn the LED on, or alternately connected such that the pin must sink current to turn the LED on. Upon system Reset, the ENC28J60 will detect how the LED is connected and begin driving the LED to the default state configured by the PHLCON register. If the LED polarity is changed while the ENC28J60 is operating, the new polarity will not be detected until the next system Reset occurs.

LEDB is unique in that the connection of the LED is automatically read on Reset and determines how to initialize the PHCON1.PDPXMD bit. If the pin sources current to illuminate the LED, the bit is cleared on Reset and the PHY defaults to half-duplex operation. If the pin sinks current to illuminate the LED, the bit is set on Reset and the PHY defaults to full-duplex operation. Figure 2-7 shows the two available options. If no LED is attached to the LEDB pin, the PDPXMD bit will reset to an indeterminate value.

FIGURE 2-7: LEDB POLARITY AND RESET CONFIGURATION OPTIONS



The LEDs can also be configured separately to control their operating polarity (on or off when active), blink rate and blink stretch interval. The options are controlled by the LACFG3:LACFG0 and LBCFG3:LBCFG0 bits. Typical values for blink stretch are listed in Table 2-1.

TABLE 2-1: LED BLINK STRETCH LENGTH

Stretch Length	Typical Stretch (ms)
TNSTRCH (normal)	40
TMSTRCH (medium)	70
TLSTRCH (long)	140

REGISTER 2-2: PHLCON: PHY MODULE LED CONTROL REGISTER

R/W-0	R/W-0	R/W-1	R/W-1	R/W-0	R/W-1	R/W-0	R/W-0
r	r	r	r	LACFG3	LACFG2	LACFG1	LACFG0
bit 15				bit 8			

R/W-0	R/W-0	R/W-1	R/W-0	R/W-0	R/W-0	R/W-1	R/W-x
LBCFG3	LBCFG2	LBCFG1	LBCFG0	LFRQ1	LFRQ0	STRCH	r
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 15-14 **Reserved:** Write as '0'bit 13-12 **Reserved:** Write as '1'bit 11-8 **LACFG3:LACFG0:** LEDA Configuration bits

1111 = Reserved

1110 = Display duplex status and collision activity (always stretched)

1101 = Display link status and transmit/receive activity (always stretched)

1100 = Display link status and receive activity (always stretched)

1011 = Blink slow

1010 = Blink fast

1001 = Off

1000 = On

0111 = Display transmit and receive activity (stretchable)

0110 = Reserved

0101 = Display duplex status

0100 = Display link status

0011 = Display collision activity (stretchable)

0010 = Display receive activity (stretchable)

0001 = Display transmit activity (stretchable)

0000 = Reserved

bit 7-4 **LBCFG3:LBCFG0:** LEDB Configuration bits

1110 = Display duplex status and collision activity (always stretched)

1101 = Display link status and transmit/receive activity (always stretched)

1100 = Display link status and receive activity (always stretched)

1011 = Blink slow

1010 = Blink fast

1001 = Off

1000 = On

0111 = Display transmit and receive activity (stretchable)

0110 = Reserved

0101 = Display duplex status

0100 = Display link status

0011 = Display collision activity (stretchable)

0010 = Display receive activity (stretchable)

0001 = Display transmit activity (stretchable)

0000 = Reserved

bit 3-2 **LFRQ1:LFRQ0:** LED Pulse Stretch Time Configuration bits (see Table 2-1)

11 = Reserved

10 = Stretch LED events by TLSTRCH

01 = Stretch LED events by TMSTRCH

00 = Stretch LED events by TNSTRCH

bit 1 **STRCH:** LED Pulse Stretching Enable bit

1 = Stretchable LED events will cause lengthened LED pulses based on LFRQ1:LFRQ0 configuration

0 = Stretchable LED events will only be displayed while they are occurring

bit 0 **Reserved:** Write as '0'

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NOTES: